VALENCE-SHELL ELECTRON-PAIR REPULSION (VSEPR) THEORY AND POLARITY

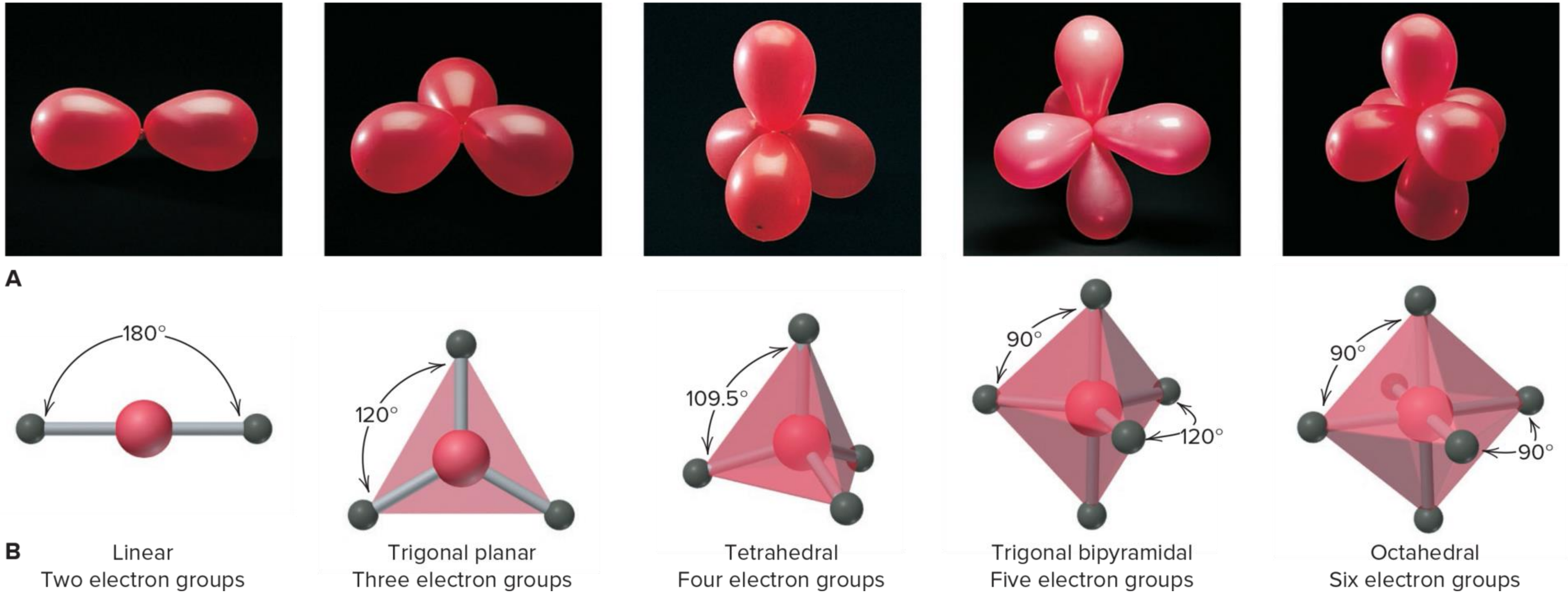
4/24/2023 and 4/25/2023

VSEPR

- Valence-Shell Electron-Repulsion Theory
  - Molecules minimize electron repulsions.
  - Each group of valence electrons around the central atom is located as far away as it can be from the others.
  - An electron group can be a lone pair, a lone electron, a single bond, a double bond, or triple bond.
  - The electron groups around the central atom affect the shape of the molecule.
  - Molecular shape is a three-dimensional arrangement of nuclei joined by bonding groups.

Molecular Shapes

- VSEPR Theory is applied to a Lewis Structure to determine the molecular shape.
- Maximum electron repulsion occurs in five different electron group arrangements.



Molecular Shape Classifications

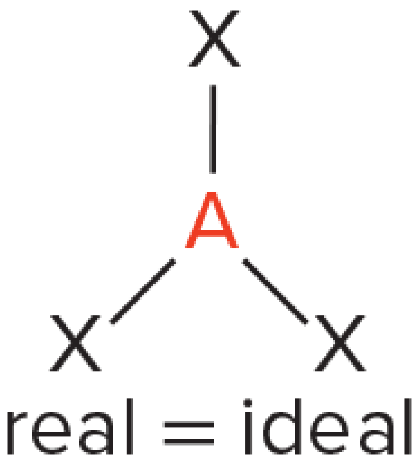
- There is a difference in the electron-group arrangement of a molecule and the molecular shape.
- Electron-group Arrangement – The bonding electron groups (shared pairs of electrons) and the non-bonding electron groups (lone pairs of electrons) around the central atoms.
- Molecular Shape – The relative nuclei that are connected only by the bonding groups of electrons.
- The image on the previous slide is an example of molecular shapes that are observed when all the electron groups around the central atom are bonding groups of electrons.

Molecular Shape Classifications

- If there are non-bonding groups around a central atom different molecular shapes are observed.
- When classifying molecular shapes an AX_mE_n designation can be used.
  - A = Central Atom
  - X = Surrounding Atom
  - E = Non-bonding Valence Electron Group
  - m and n are integers

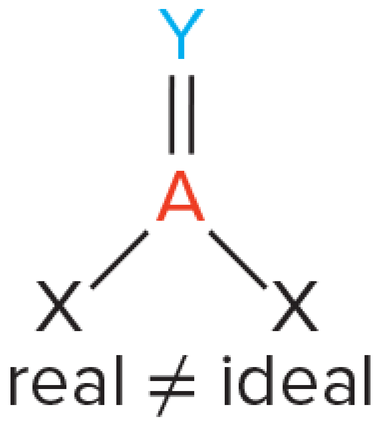
Bond Angle

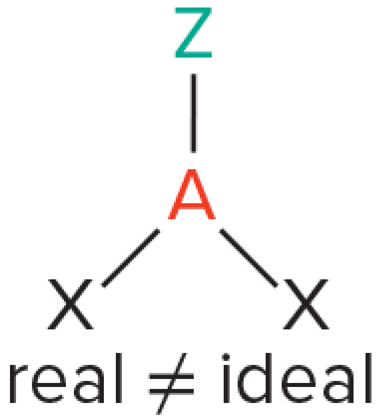
- Formed when bonds of joining nuclei of two surrounding atoms to the central atom's nucleus.
- Ideal bond angles (the bond angles on slide 3) are determined using geometry.
  - Observed when all bonding groups are the same and are connected the same type of atom.

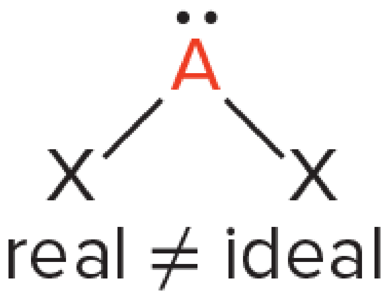


Bond Angle

- Real bond angles deviate from ideal bond angles when the bonding groups are not the same or connected to the central atom differently.

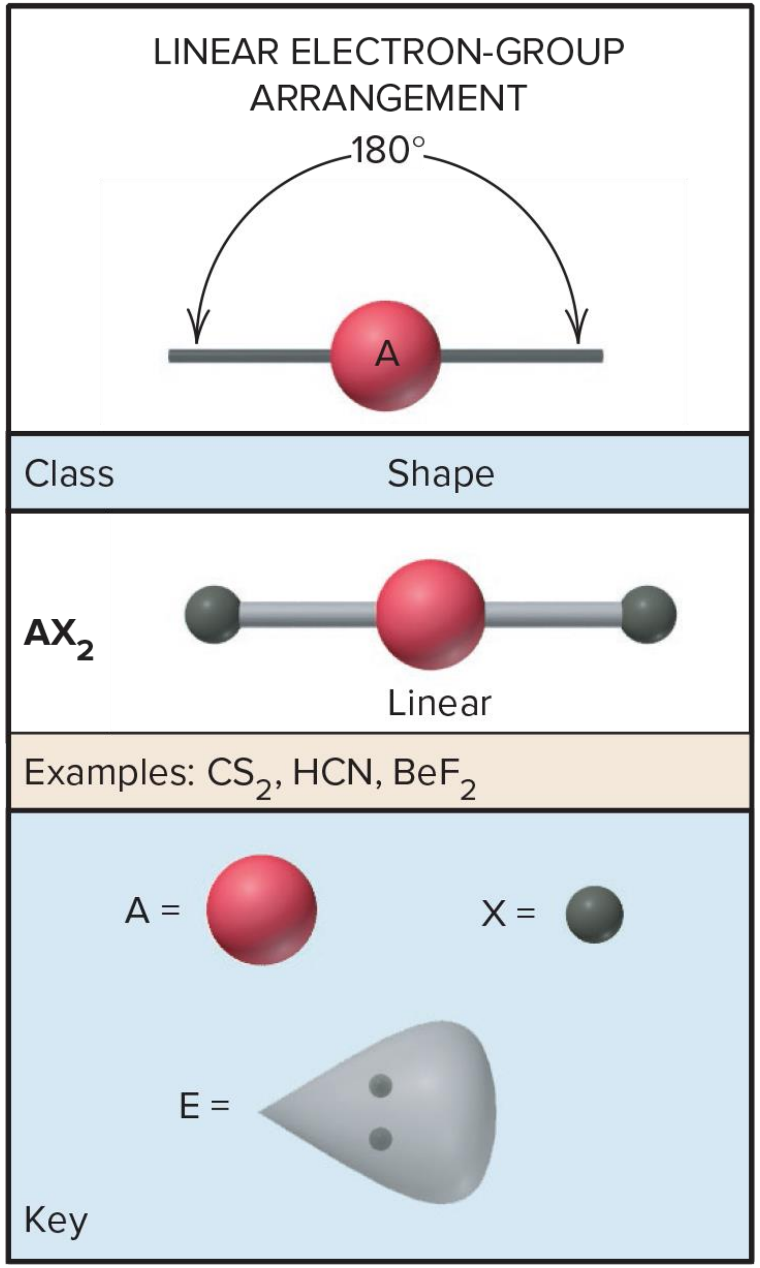






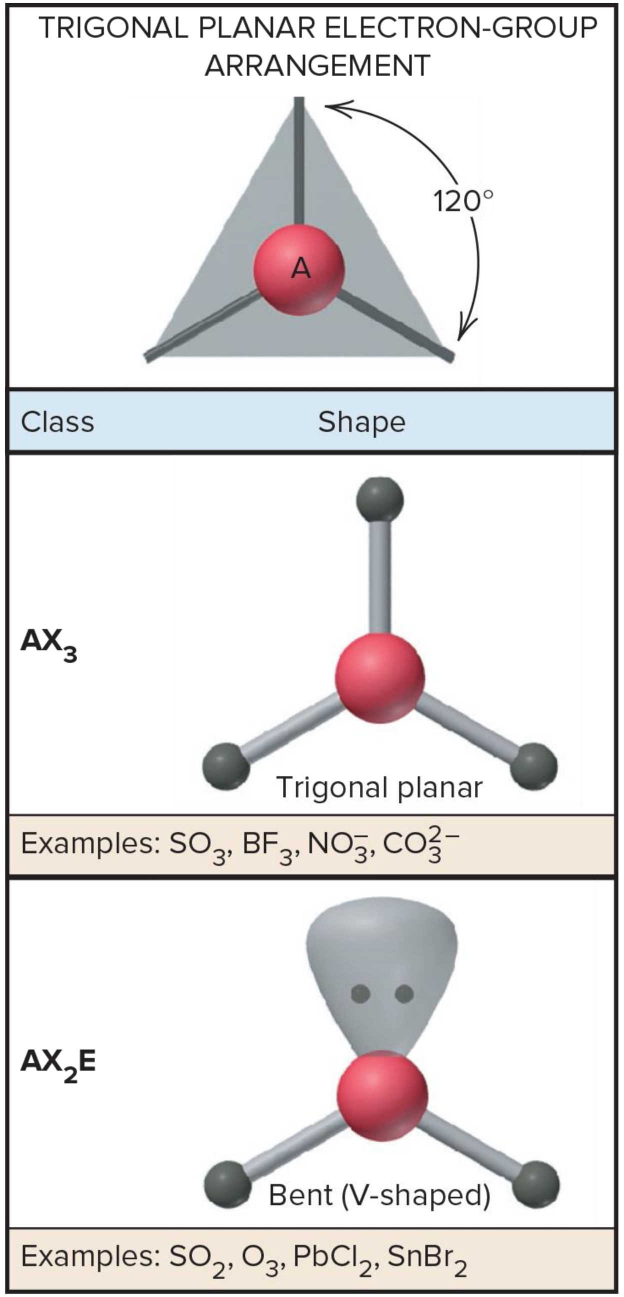
Linear Electron Group Arrangement

- Two electron groups are connected to a central atom in opposite directions.
- Molecular shape is linear and the bond angle of 180°.



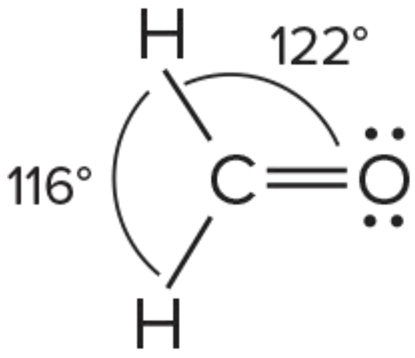
Trigonal Planar Electron Group Arrangement

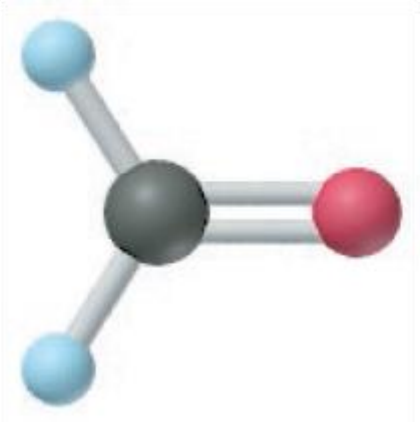
- Three electron groups are connected to a central atom pointing to corners of an equilateral triangle.
- Bond Angles 120°.
- Molecular shapes:
    1. All bonding groups are present.
    2. One bonding group contains a double bond.
    3. A lone pair is present.



How a Double Bond Effects Bond Angles

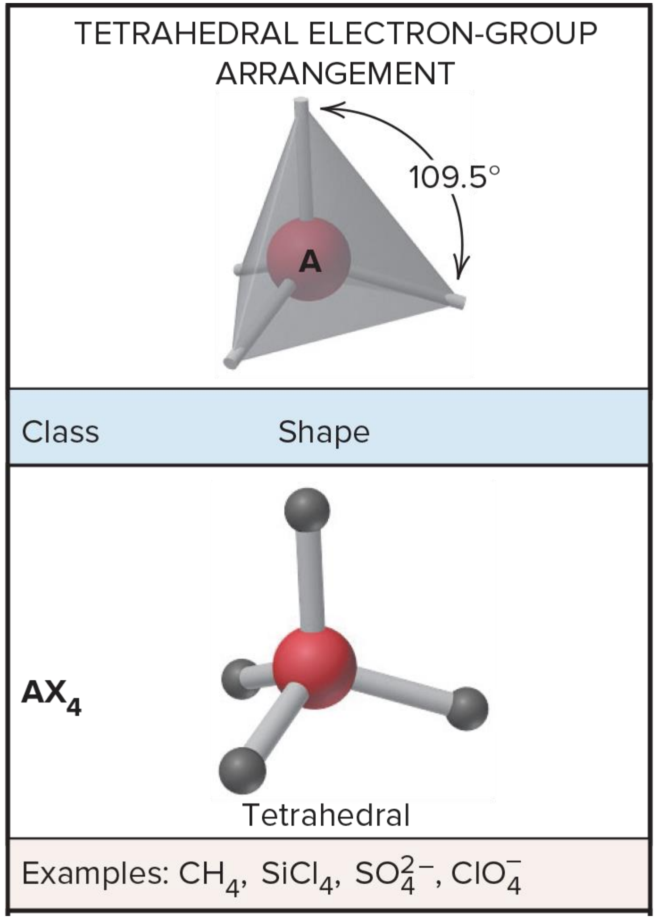
- The double bond has more electron density than the single bonds.
- Repels the electrons in a single bond more than a single bond repels a single bond.
- Ex.
  - Formaldehyde

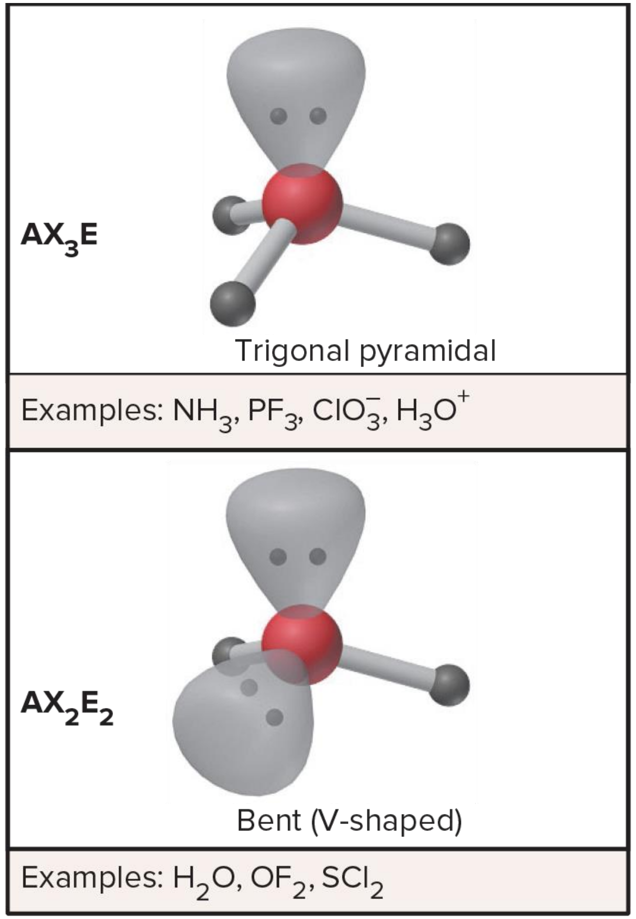




Tetrahedral Electron Group Arrangement

- Four electron groups are connected to a central atom pointing to corners of an tetrahedron.
- Polyhedron in which four faces are made of equilateral triangles.
- Bond angle is 109.5°.

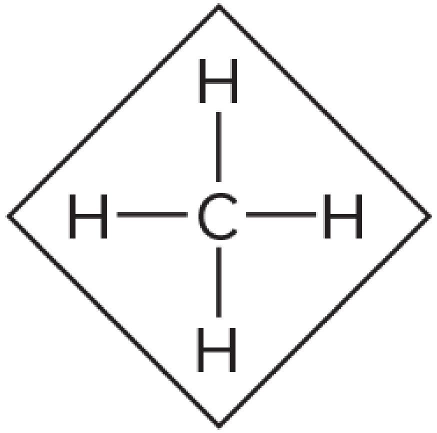


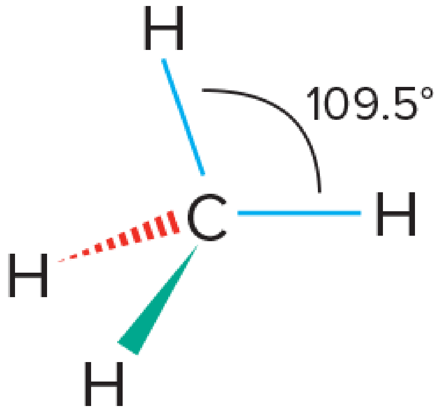


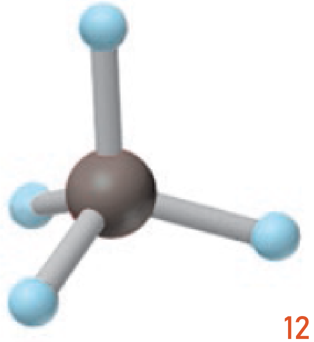
Tetrahedral Electron Group Arrangement

- All molecules and ions with four electron groups will have the tetrahedral arrangement.

1. All Bonding Groups (AX_4)– Common electron geometry for organic molecules; Tetrahedral Molecular Shape.
   - Ex.
     - Methane



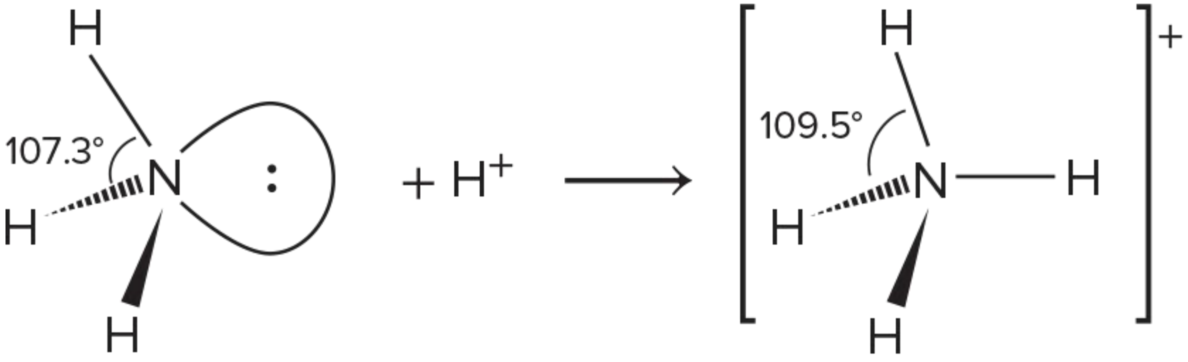


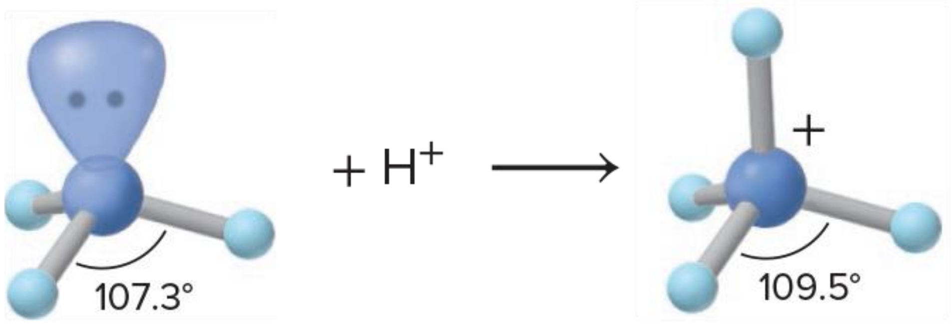


Tetrahedral Electron Group Arrangement

2. One Lone Pair (AX_3E) – Has a single lone pair of electrons with stronger repulsions to other electron groups.
   - Bond angle of bonding groups if 107.3°; Trigonal Pyramidal Molecular Shape.

- Ex.
  - Ammonia





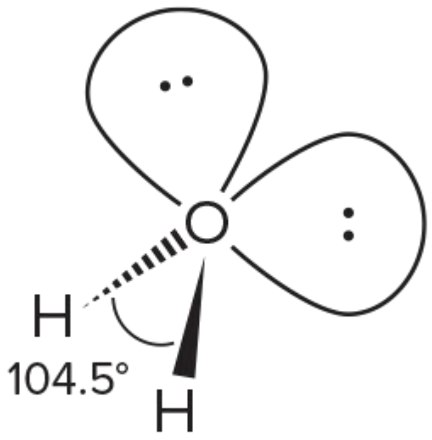
Trigonal pyramidal

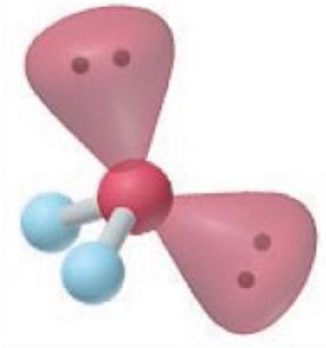
Tetrahedral

Tetrahedral Electron Group Arrangement

3. Two Lone Pairs (AX_2E_2) - Has two lone pair of electrons with stronger repulsions than a single lone pair of electrons.
   - Bond angle of bonding groups if 104.5°; Bent Molecular Shape (V-Shaped Molecule).

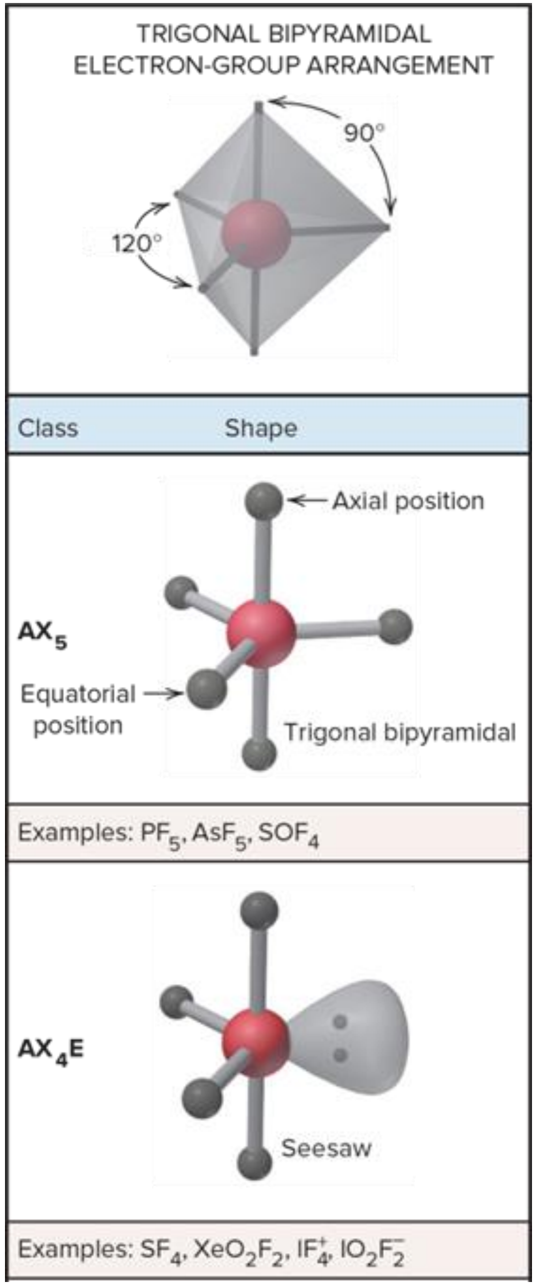
- Ex.
  - Water

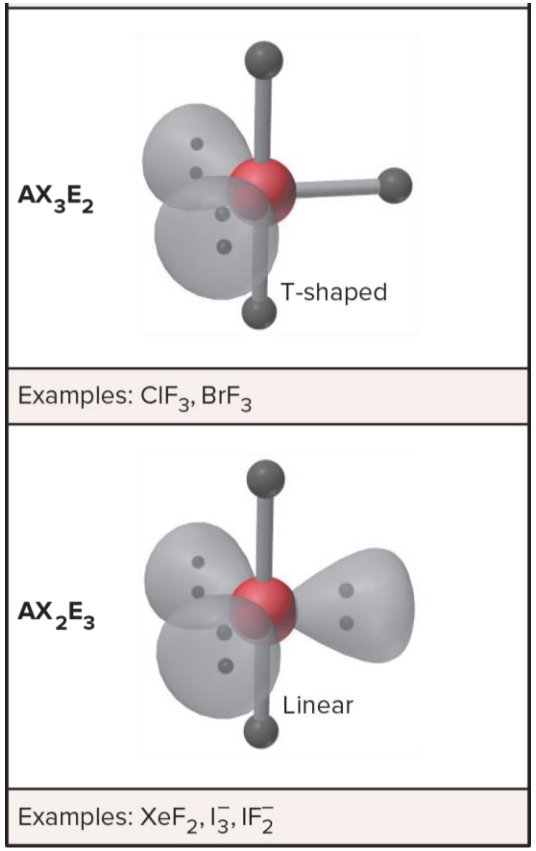




Trigonal Bipyramidal Electron Group Arrangement

- Five electron groups all repelling each other.
- Two trigonal Pyramids sharing a common base.
  - Equatorial Groups (3)– Reside on trigonal plane; 120°.
  - Axial Groups (2) – Reside above and below the trigonal plane; separated from the equatorial groups by a 90° angle.





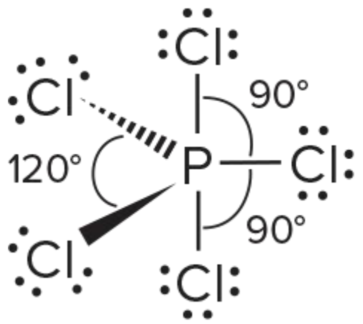
Trigonal Bipyramidal Electron Group Arrangement

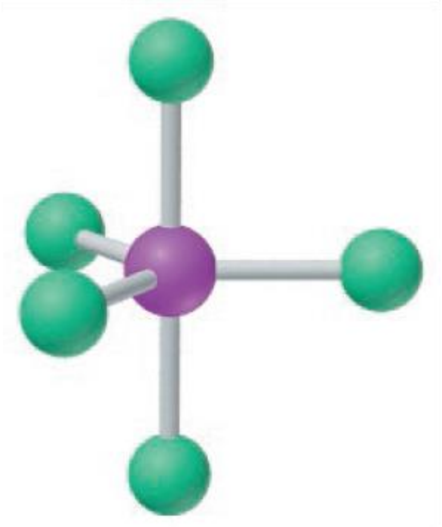
- Electron repulsion strength affects where lone pairs reside.
- Equatorial – equatorial (120°) electron repulsions are weaker than axial-equatorial (90°) electron repulsions.
- Due to lone pairs having stronger repulsions lone pairs occupy equatorial positions when possible.

Trigonal Bipyramidal Electron Group Arrangement

- All Bonding Groups (AX_5) – Trigonal Bipyramidal Shape.
  - Five identical surrounding atoms has ideal bond angles.

- Ex.
  - Phosphorous Pentachloride

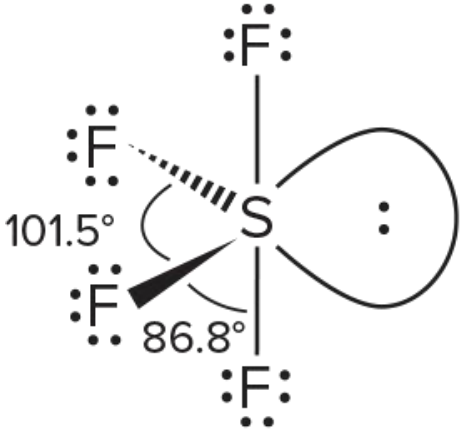


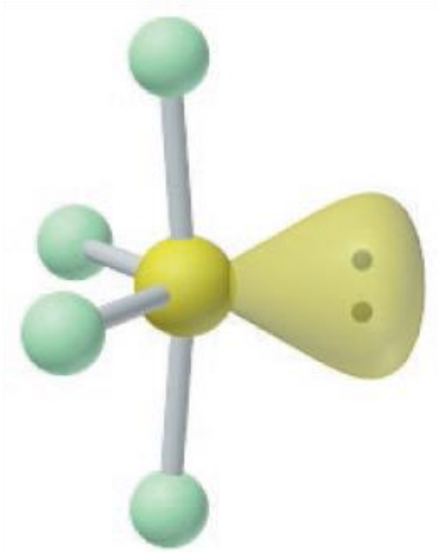


Trigonal Bipyramidal Electron Group Arrangement

2. One Lone Pair (AX_4E) – Seesaw molecular shape.
   - The lone pair is in an equatorial position minimizing electron repulsions.
   - Bond angles are 101.5° and 86.8°.

- Ex.
  - Sulfur tetrafluoride

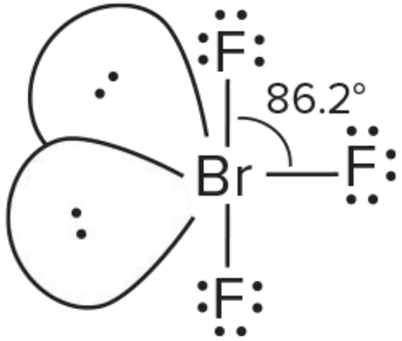


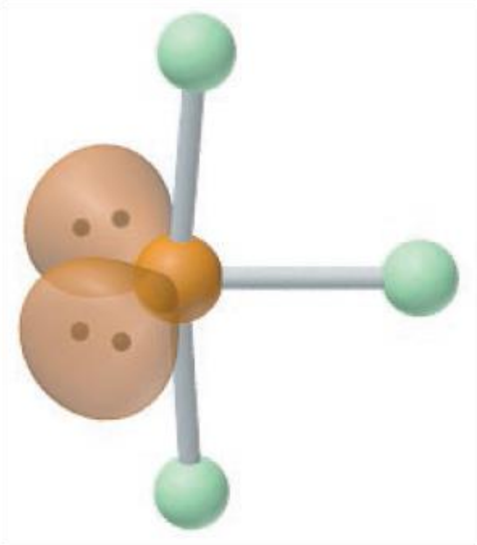


Trigonal Bipyramidal Electron Group Arrangement

3. Two Lone Pairs (AX_3E_2) – T-shaped molecule.
   - Both lone pairs are in equatorial position.
   - Axial-equatorial bond angle reduced to 86.2°.

- Ex.
  - Bromine trifluoride

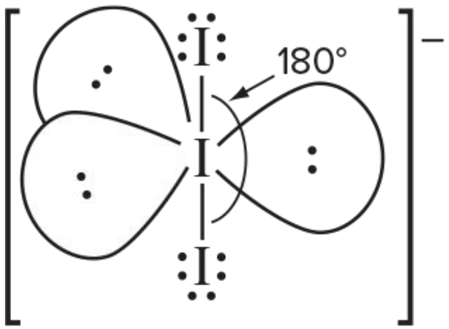


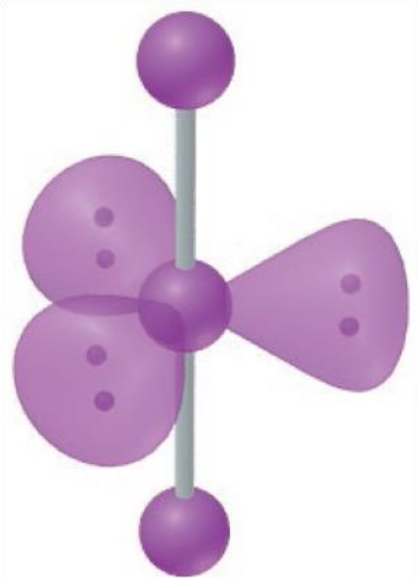


Trigonal Bipyramidal Electron Group Arrangement

4. Three Lone Pairs (AX_2E_3) – Linear molecular shape.
   - Three equatorial lone pairs give the two axial boning pairs a 180° bond angle.

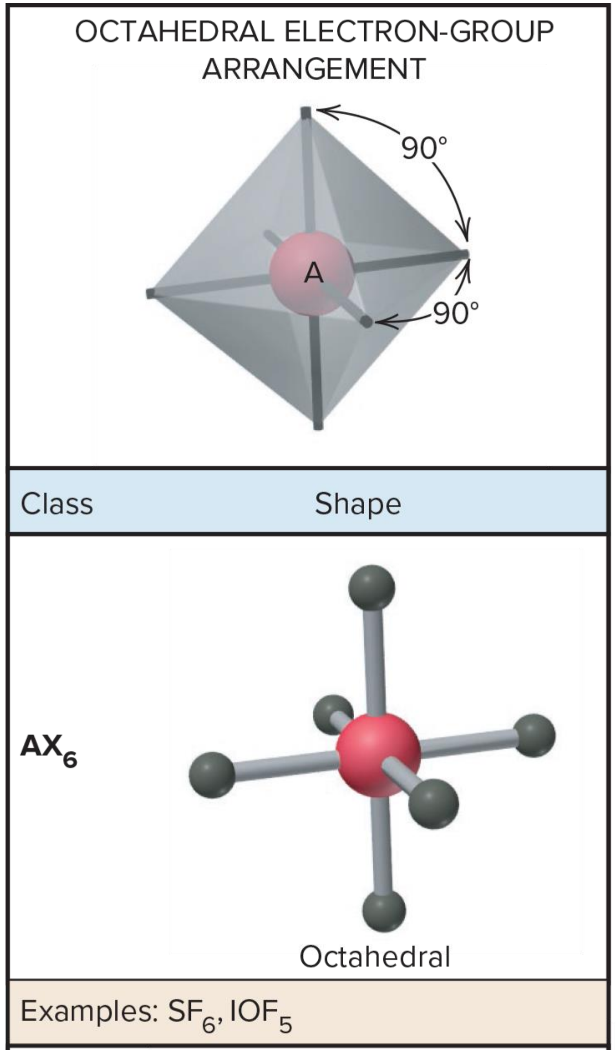
- Ex.
  - Triiodide ion

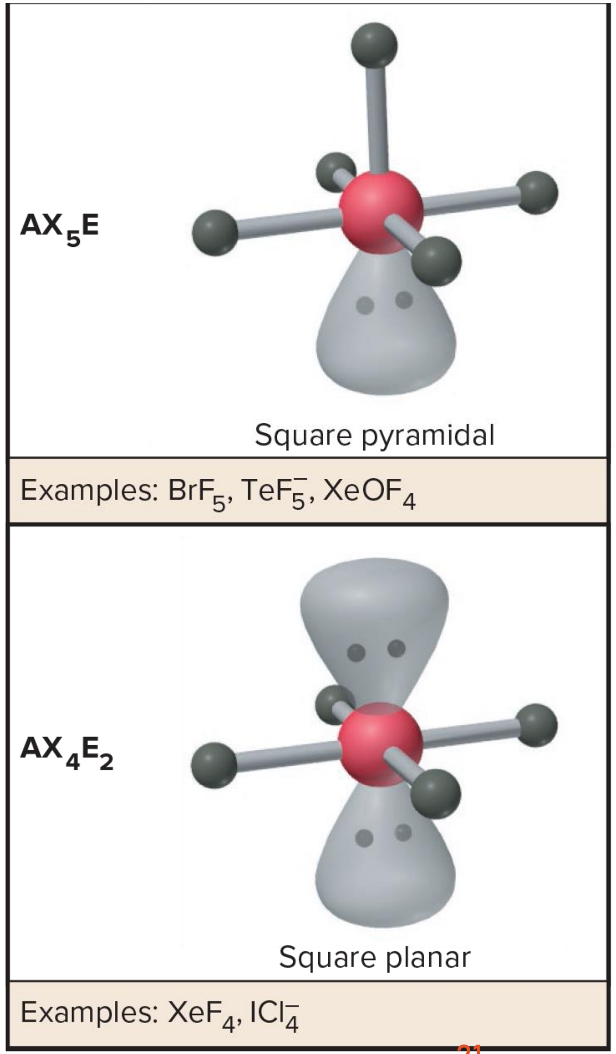




Octahedral Electron Group Arrangement

- Six electron groups all repelling each other.
  - Polyhedron made of eight equilateral triangles and six identical vertices.
  - Ideal bond angles are 90°.

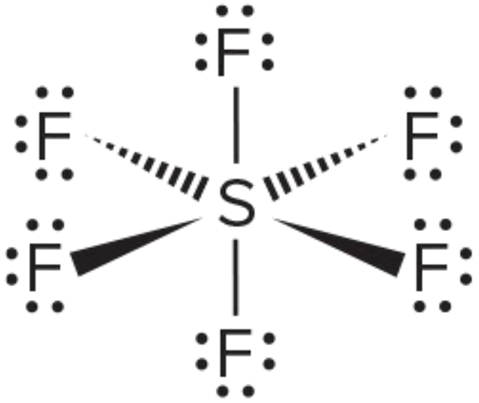


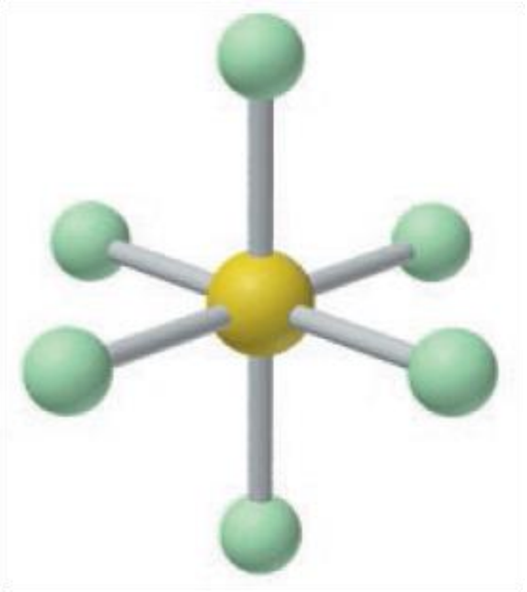


Octahedral Electron Group Arrangement

1. All Bonding Groups (AX_6) – Octahedral Shape
   - Bond Angles 90°.

- Ex.
  - Sulfur hexafluoride

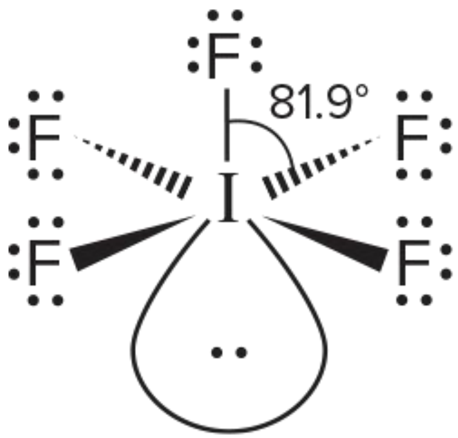


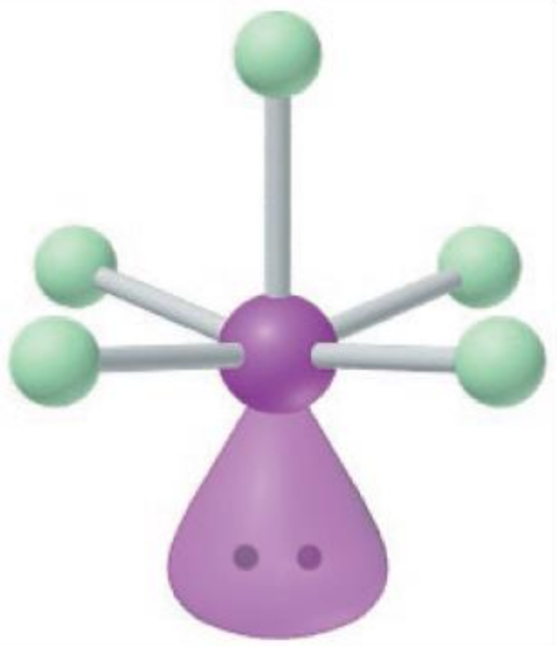


Octahedral Electron Group Arrangement

2. One Lone Pair (AX_5E) – Square pyramidal molecular shape.
   - Lone Pair causes bond angles to reduce to 81.9°.

- Ex.
  - Iodine pentafluoride

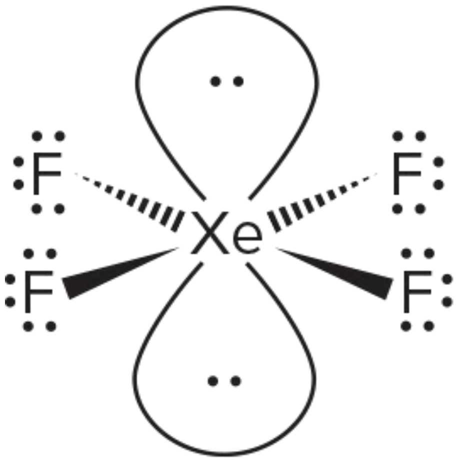


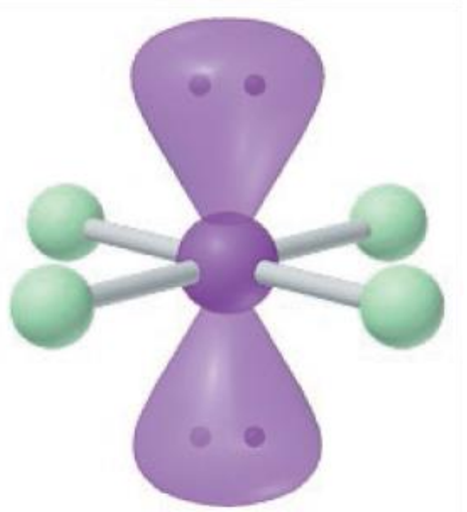


Octahedral Electron Group Arrangement

3. Two Lone Pairs (AX_4E_2) – Square planar molecular shape.
   - Lone pairs are opposite each other to reduce lone pair – lone par repulsions.
   - Bond angles remain 90° since each lone pair reduces the bond angle by the same magnitude.

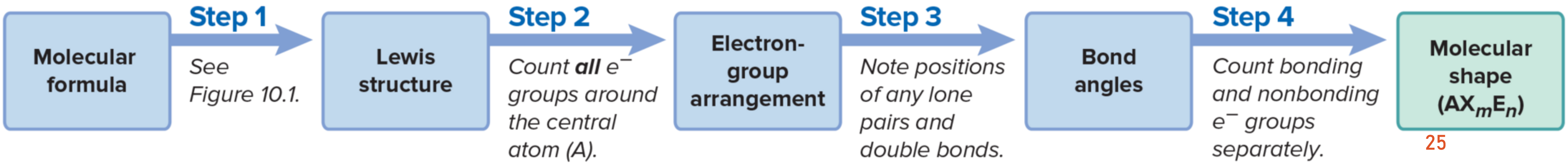
- Ex.
  - Xenon tetrafluoride





Using VSEPR to Determine Molecular Shape

1. Write the Lewis Structure of the molecular formula.
   - Can see the number of electron groups around central atom and relative atom placement.
2. Assign one of the five electron-group arrangements.
   - Count all electron groups around the central atom.
3. Predict the ideal bond angle and how any lone pairs or double bonds may cause a deviation in bond angles.
4. Draw the name and molecular shape by counting bonding and non-bonding electron groups.

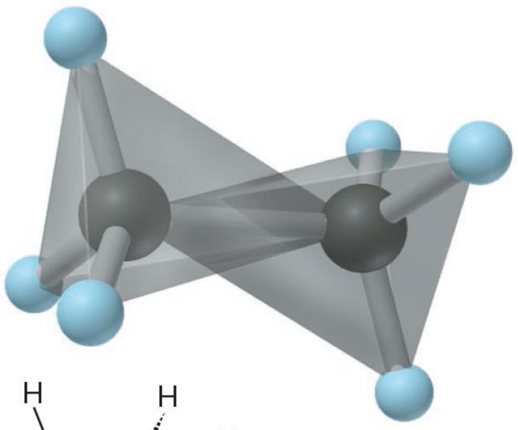


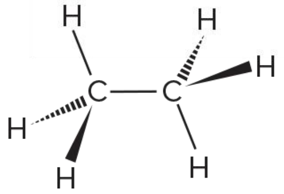
Using VSEPR to Determine Molecular Shape

- Ex.
  - Draw the molecular shape and predict the bond angle (relative to the ideal bond angle) of SOF_4.

Molecular Shapes Containing more than one Central Atom

- A molecule containing more than one central atom is a composite of the shapes around each central atom.
- Ex.
  - Ethane – has two central carbon atoms with no lone pairs.
  - Shaped as two tetrahedra that overlap.

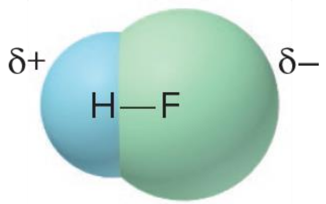




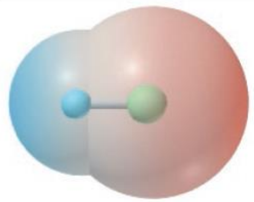
Polarity and Molecular Shape

- The polarity of a molecule influences a molecules melting point, boiling point, solubility, reactivity, and biological function.
- Covalent bonds containing atoms that unequally share electrons are polar.
- In a large molecule the shape of the molecule and bond polarity determine the molecules polarity.
- A molecule is polar when there is an uneven distribution of charge over the whole molecule or most of the molecule.
- Polar molecules can be oriented in an electric field in which the partially charged ends are attracted to the oppositely charged magnetic plates.
- Molecular polarity is measured by a dipole moment.
  - Unit: debye (D) $1\ D = 3.34 \times 10^{-30}\ C \cdot m$

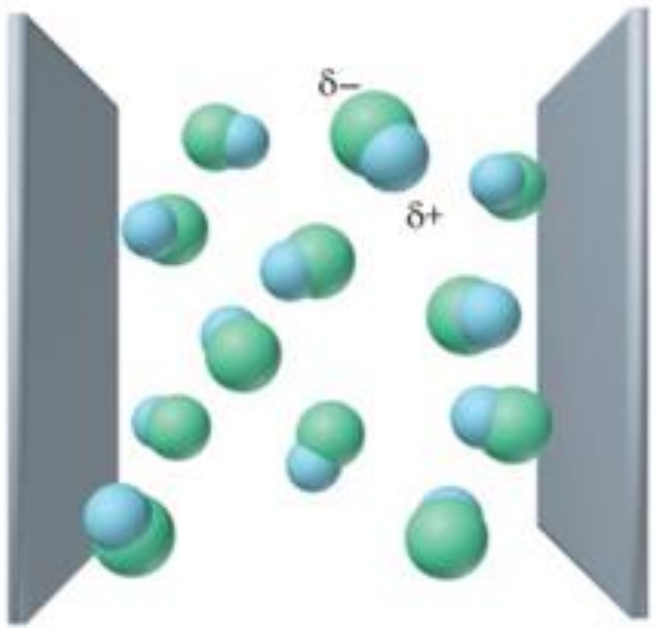
Polarity and Molecular Shape



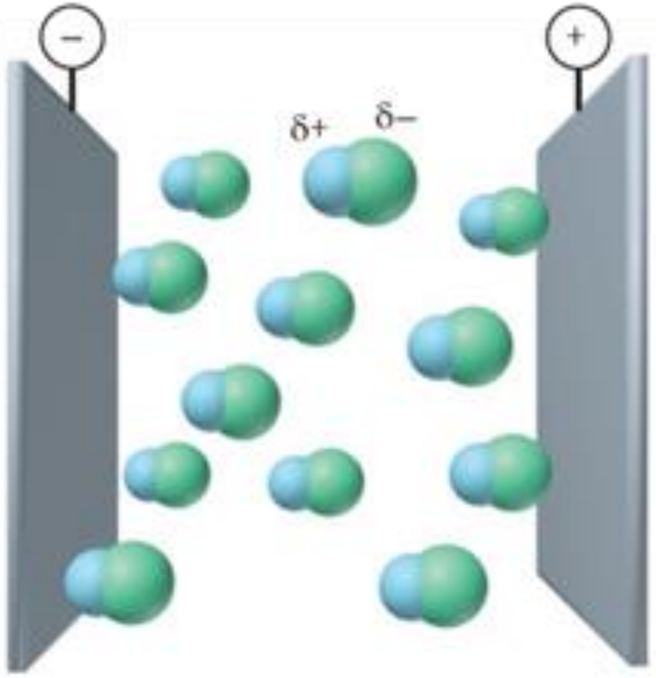
Space-filling model of HF



Electron density model of HF



Electric field off: *HF molecules are oriented randomly.*



Electric field on: *HF molecules are oriented with their partially charged ends toward the oppositely charged plates.*

Bond Polarity, Angle, and Dipole Moment

- A molecule is not always polar though it may contain polar bonds.
- The shape of the molecule and atoms around the central atom must be considered.
- Three examples:

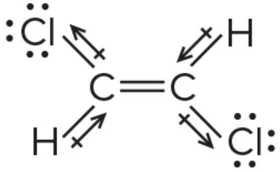
1. CO_2
   - C EN = 2.5; O EN = 3.5
2. H_2O
   - H EN = 2.1; O EN = 3.5
3. Molecules with the same shape and different polarities.
   - CCl_4 and $CHCl_3$
     - Cl EN = 3.0

Predicting Molecular Polarity

- Ex.
  - Predict the bond direction of the dipole and molecular polarity of:

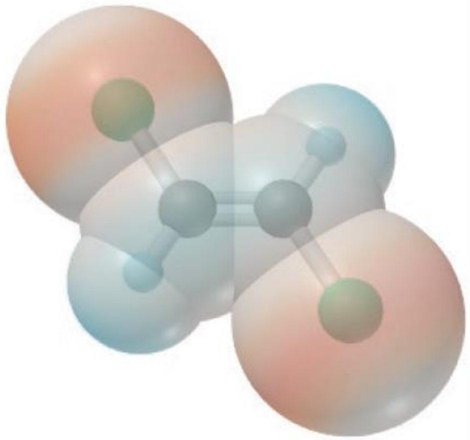
1. Ammonia
2. Boron trifluoride
3. Carbonyl sulfide

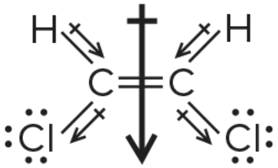
How Molecular Polarity Effects Molecular Behavior



trans

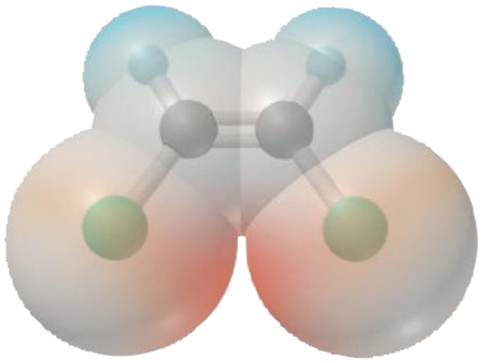
$\mu = 0$ D
Boiling point = 47.5°C





cis

$\mu = 1.90$ D
Boiling point = 60.3°C



How Molecular Polarity Effects Molecular Behavior

- Atomic property influences on a substance's behavior.